Exercise courses

Start with Exercises 1 and 2. Then choose an exercise 3 or 4 without assistance.
to 7 are a reserve of transfer, research or programming: the professor designates one.
Exercise 8 resumes at a distance.

Exercise 1.1 · Taking charge of indices

The following is considered (u_n) defined for all natural n by $u_n = 2n^2 - 3n$. The first term is therefore the level 0.

1. Calculate separately u_0, u_1 and u_4, by showing the substitution of n.

2. Express u_{n+1} according to the n, then develop and reduce this expression.

3. Calculate $u_{n+1} - u_n$. By studying his sign for the integer ranks, describe the variations in the sequence.

Exercise 1.2 · Choosing the right model

A nursery now has 100 young plants, and is considering two models to predict its future.
effective: in model A, the number increases by 15 plants per month; in model B, it increases
10 per cent per month. a_n and b_n the anticipated number of staff after n month, with $a_0 = b_0 = 100$.
Two models are studied separately.

1. **Model A.** Write the relationship between a_{n+1} and a_n. Recognize the nature of the sequence, then express a_n according to the n. Calculate a_2.

2. **Model B, after the study of geometric sequences.** Resuming the same questions for (b_n) and calculate b_2.

3. For each model, explain what is assumed to be constant.
A very long-term forecast could become unrealistic.

Exercise 1.3 · Find a sequence

The rest (u_n) is geometric, defined from row 0. q It's strictly positive.
knows only $u_1 = 6$ and $u_3 = 54$.

1. Determine q by linking the two known terms. Present all the steps that allow choose its value.

2. Deduct from it $u_0$then u_5.

3. What would change the deletion of the hypothesis? $q > 0$? Give all possible reasons and explain whether the value u_5 would be amended.

Exercise 1.4 · Seat rows

A showroom has 12 rows. The first row has 18 seats; each row has a seat.
next has two more seats than the previous one. u_n the number of seats in the
row n, for 1 $\leq n \leq 12$.

1. Define the sequence (u_n) and calculate the number of seats in the last row.

2. Calculate the total capacity of the room. Explain the number of terms retained in the sum.

3. Calculate the average number of seats per row and use this result to control the total found.

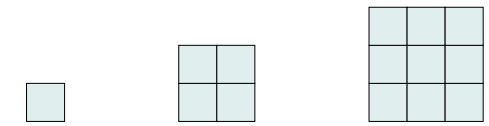
Exercise 1.5 · Accumulate squares

Patterns are built with identical square tiles. The row 1 pattern contains a tile.
To pass from the rank pattern n at the rank level n + 1, add 2 n + 1 tiles. u_n the
number of tiles in the row pattern n, for $n \geq 1$.

1. Calculate u_2 and u_3. Draw a layout of these tiles that suggests a formula for u_n.

2. Conjecture an expression of u_n according to the n.

3. Verify that this expression gives the first term and satisfies the construction rule.
why these two audits identify the follow-up.



Rank 1 Rank 2 Rank 3

A possible provision for the first three reasons; the drawing helps to formulate a conjecture, which
We have to justify that.

Fiscal year 1.6 · Two refunds

A person must repay 600 monetary units, without interest. Two plans are envisaged.
Plan A, it pays 50 units at the end of each month until repayment.
The calculation of model B is carried out without
Rounded. a_n and b_n the remaining debts after the payment of the month n, $a_0 = b_0 = 600$.

1. Express a_n until the end of the debt. At what month is the repayment completed?

2. Express b_n. Determine the first month in which the remaining debt is strictly less than 10 units
; Check the previous month.

3. In Model B without rounding, does the debt become exactly zero after a finite number of
month ? Justify and comment on the difference with a real situation.

Exercise 1.7 · Read a list

A program must retain the terms of the sequence defined by $u_0 = 2$ and $u_{n+1} = 3u_n - 1$. It
starts with the instructions `u=2` and `L=[u]`. He then executes five times, in this order, `u=3*u-1`
then `L.append(u)`.

1. Run the first two passages from the loop by hand. Then give the contents of the list.
2. How many elements contains the list at the end of the five passages? What is the rank of the last term recorded?
3. Write the complete program in Python, then check the last item value.
   why the number of terms exceeds the number of passages by one unit.

Fiscal year 1.8 · Deferred review

This question is dealt with one week after the chapter, without consulting the forms.
defined for $n \geq 2$ by $u_2 = 12$ and $u_{n+1} = u_n - 3$.

1. Calculate u_7 specifying how many passages separate rows 2 and 7.
2. Calculate the sum $u_2 + u_3 + \cdots + u_7$. Explain the difference between the number of terms of this sum and the number of passages calculated on the first question.

Personal work. Perform only the task assigned at the end of the meeting, in progress
Date: 10·20 min maximum. The exercises depend on the concepts already studied; some questions
Resuming exercise 8 a week after the chapter (10 min).
is exceeded, note the difficulty and bring the tests; do not accumulate the whole reserve.

To be done in the CQFD manual

These exercises are dealt with in your manual. The progress indicates the session; they replace the personal work planned that day (10 to 20 minutes in total).

CQFD, exercise 9 p. 64, questions a and b. Calculate the first four terms.
substitutions before simplifying.

CQFD, exercise 11 p. 64, all questions. Calculate the terms successively and keep
the trace of the four stages.